

Clinical Knowledge Assistant: HIPAA Compliance & Clinical Safety Audit

1. HIPAA & Regulatory Compliance Matrix

HIPAA Security Rule Standard	Implementation Detail	Compliance Status
Transmission Security	TLS 1.3 encryption in transit for all queries and vector payloads	COMPLIANT
Encryption at Rest	AES-256 encrypted vector database and persistent storage	COMPLIANT
Access Control & Audit Logs	Role-based access control (RBAC) with immutable query audit logs	COMPLIANT
De-Identification Standard	18 HIPAA identifiers stripped before embedding generation	COMPLIANT
Business Associate Agreement	Verified active BAA with API infrastructure providers	COMPLIANT

2. Clinical Grounding & Hallucination Audit

- 100 clinical test questions evaluated by medical subject matter experts.
- 0% fabricated drug dosages or contraindicated treatment recommendations.
- Strict grounding prompt successfully rejected out-of-domain queries.